THE RELATIONSHIP BETWEEN GENTRIFICATION AND CRIME IN AUSTIN, TEXAS FROM 2003 TO 2013

Overview

The Austin, Texas Metropolitan Area (MSA) population more than doubled between the years 1990 and 2010, from 846,227 to 1,716,289 people. Concern is growing over how this increase in population, and accompanying economic growth, have affected housing affordability in Austin. Specifically, the increase in property values and the subsequent increase in tax valuations in working class neighborhoods of Austin have driven a number of moderate to low income residents out of the area.

Previous studies into the relationship between gentrification and crime have had mixed results with crime depending on gentrification as having a positive relationship, negative relationship, curvilinear relationship or even showing that gentrification depends on crime.

This study analyzes the relationship between gentrification and crime in Austin, Texas for U.S. census tracts determined to be gentrified between 2003 and 2013. With 24 gentrified and 24 non-gentrified tracts in the study, crime was split into three groups: property crime, violent crime, and non-indexed crime. For each type of crime, the change in annual crime per 1000 people between the years 2003 and 2013 was calculated for gentrified, non-gentrified, and tracts belonging to the rest of Austin not included in the analysis.

Criteria for Choosing Gentrified and Non-Gentrified Tracts

In this study, a census tract must pass the following two tests in ordered to termed “gentrified”. The criteria for test 1 and test 2 help ensure that the tracts labeled as "gentrified" in this study are not counted based on normal increases in education, property values and family income and to ensure that only tracts that were considered in the study were under average for all three criteria at the beginning of the study. From the Gentrification Report Methodology section in the Governing website, tracts labeled as "not eligible" either had missing data or did not meet any one of the measures in test 1 or test 2 stated above. For this reason, "not eligible" tracts were left out of the study

First Test

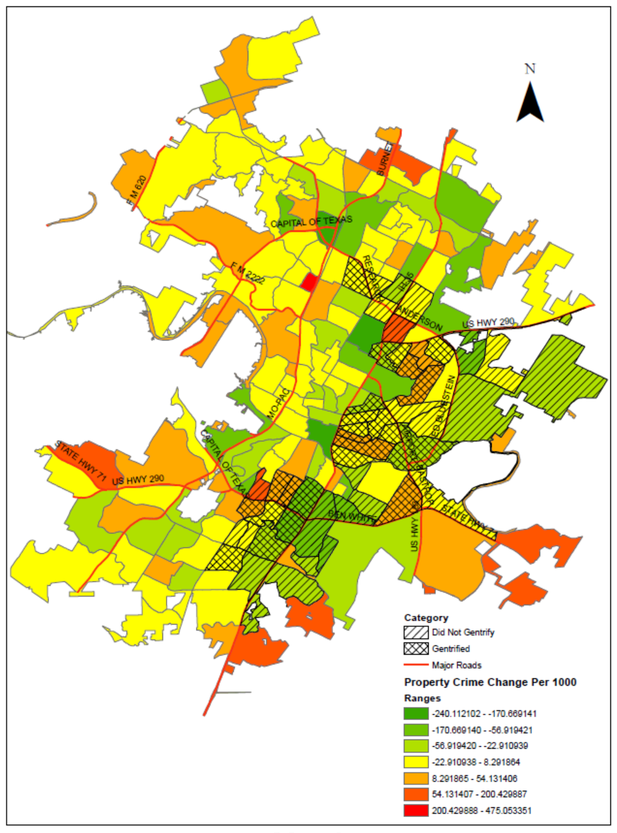
At the beginning of the period of study, to qualify, a tract will need:

1) a population of at least 500 people
2) The tract's median household income needs to be in the bottom 40th percentile for the metro area it's located in
3) The tracts median home value will need to be in the bottom 40th percentile for the metro area it's located in.

Second Test (Note: Must Pass First Test First)

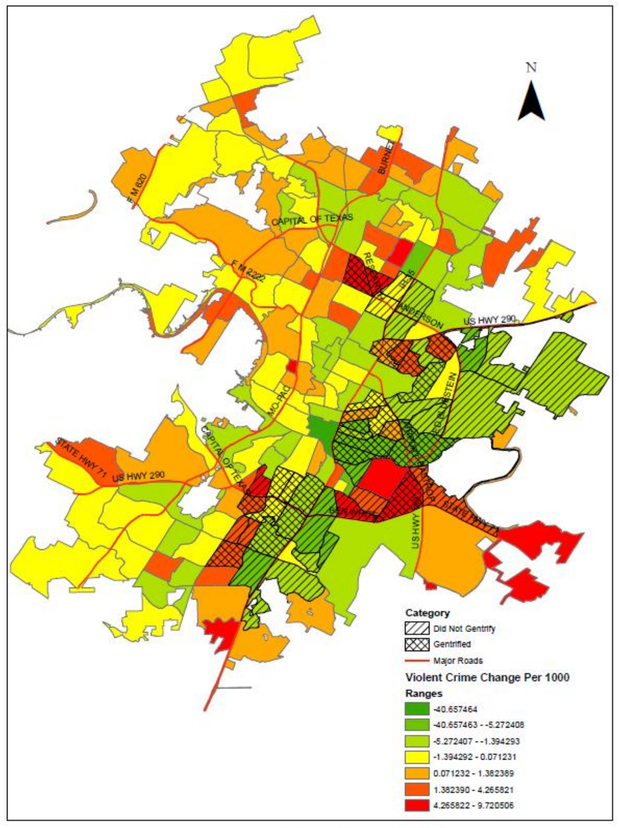
Gentrification took place if the following three are met after the 10 year study

1) the percentage of residents with a bachelor's degree age 25 and older will need to be in the top third percentile for the metro area
2) The median home value will have increased for a tract when adjusted for inflation
3) The percentage increase of inflation adjusted median home value is in the top third percentile for the metro area.



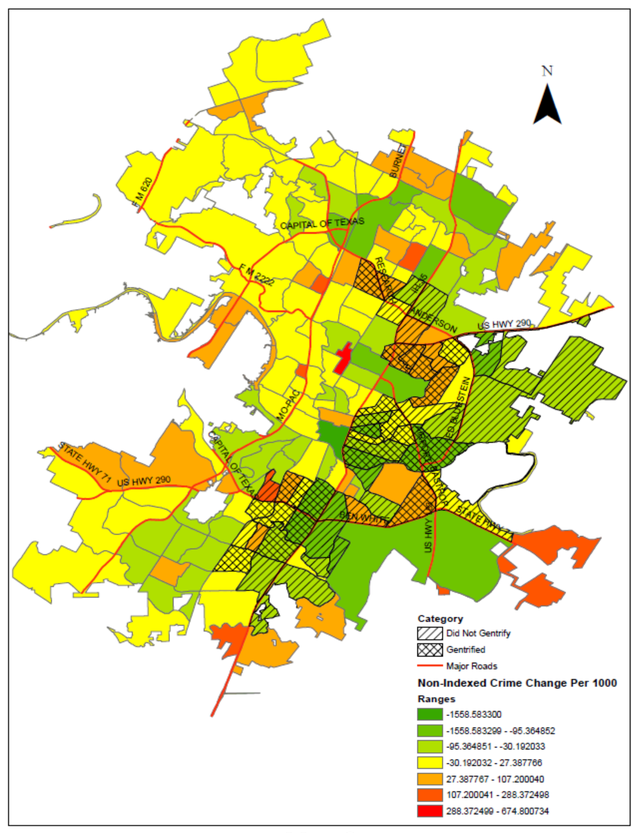
Map 1

Annual Property Crime Change 2003-2013 per 1000, Austin, Texas



Map 2

Annual Violent Crime Change 2003-2013 per 1000, Austin, Texas



Map 3

Annual Non-Indexed Crime Change 2003-2013 per 1000, Austin, Texas

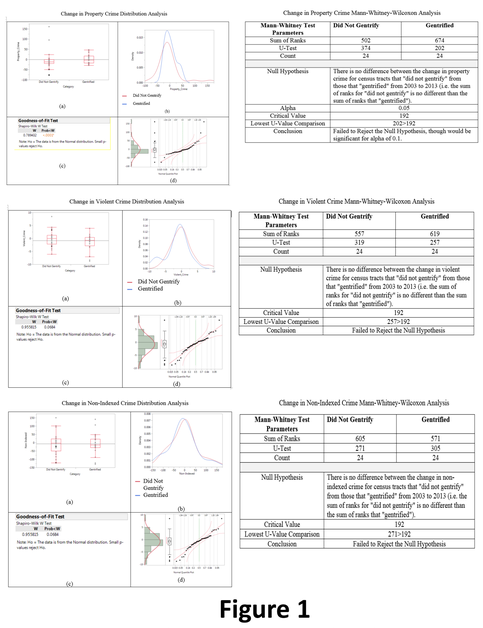


Figure 1

Distribution and Statistical Analysis

Indexed and Non-Indexed Crimes

Crime can be divided into "indexed" and "non-indexed" crime. Examples of indexed crime are: murder, rape, robbery, aggravated assault, burglary, theft, motor vehicle theft, and arson. "Non-Indexed" crime includes most other crimes and includes "simple" assault, forgery, weapons violations, drugs, criminal mischief, public intoxication, disorderly conduct, and various city ordinance violations among many others. Property, violent and non-indexed crime are analyzed in this study. This study defines “violent crime” as a combination of robbery and aggravated assault.

Results / Conclusion

- No statistically significant difference in the change in crime between gentrified and non-gentrified tracts.
- Overall, the results counter the perception that gentrified tracts are somehow safer than non-gentrified tracts and it might not be a valid reason to justify activities by government entities or investors aiming to develop economically depressed neighborhoods at risk for gentrification and help them make smarter choices in the planning to prevent this from happening.
- This kind of analysis can be helpful in quantifying the effects of gentrification.
- The results from these studies may give local governments and investors better tools for their decision process on where and how to spend tax dollars and investment money respectively.

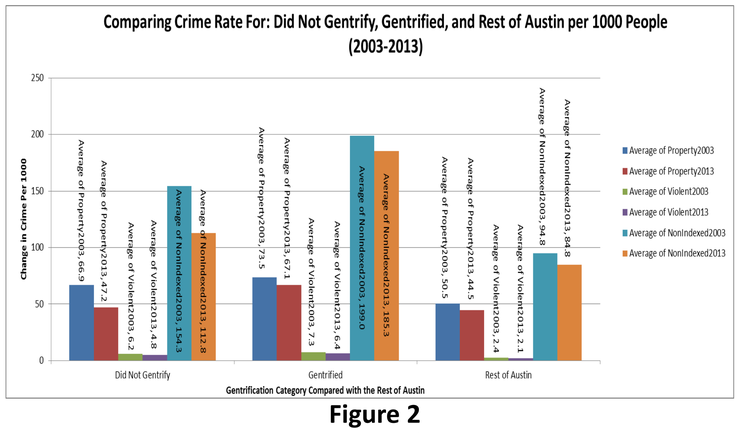
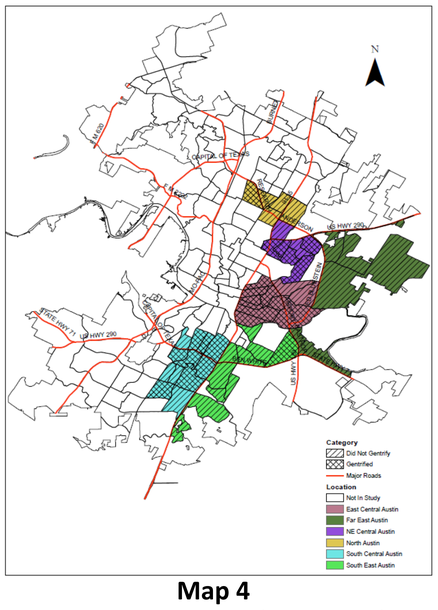


Figure 2

Comparing Crime Rate For: Did Not Gentrify, Gentrified, and Rest of Austin per 1000 People (2003-2013)



Map 4

2000 Census Tracts in Spatial Variation Study

Spatial Variation of Crime and Gentrification

This study also performed a cursory analysis by dividing all 48 gentrified and non-gentrified tracts into six parts based on location in the city. This analysis revealed a significant difference in the average change in crime between gentrified and non-gentrified tracts based on where the tracts are located in the city. The spatial variation for the rate of change in crime for property, violent and non-indexed crime in the city is shown in Map 4. Note that the "Far East Austin" and the "Rest of Austin" sections have no gentrified tracts to compare to. One example is the dramatic increase in all three types of crime for the “Gentrified” tracts labeled as “South East Austin” for this study compared with “Did Not Gentrify”. Another example is the dramatic decrease in non-indexed crime for “Gentrified” tracts labeled as “East Central Austin” compared with “Did Not Gentrify”. The opposite is true for the tracts labeled as “North Austin” where the non-indexed crime dramatically increased for the tracts labeled as “Gentrified”. Note: Unlike the study above, the separate areas here have an uneven number of “Gentrified” and “Did Not Gentrify” tracts negatively affecting this analysis.

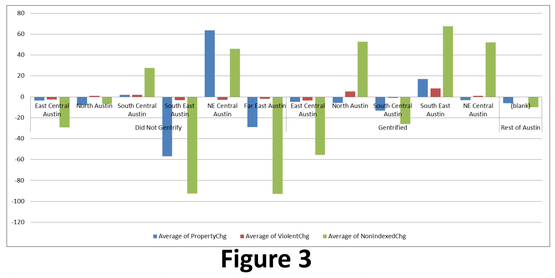


Figure 3

Spatial Variation for: Did Not Gentrify, Gentrified, and Rest of Austin per 1000 People (2003-2013)

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- Data Source: Austin Police Department
- Census Tract Shape File Source: Capital Area Council of Governments (CAPCOG)
- GIS Software: ArcMap 10.2.2
- Statistical Analysis Software: SAS JMP 12
- From the Directed Research Paper for Texas State University in partial fulfillment of the Master of Applied Geography (MAG)
- The complete study along with all of the references can be found at: http://www.henryreissig.com/directed-research-project

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